Read each of the questions carefully and answer showing all steps of your solution

Final Answers with no clear solution get ZERO

All the computations must be rounded to exactly four decimal places

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Q1- Find the root of the function: $f(x) = 2e^{\cos(x)} - e^{x^2} + 1$

15

Use the **Newton Raphson Method**. Perform **three iterations** to find the root, assume $x_0 = 1$

(15 marks)

| $n$ | $x_n$ | $f(x_n)$ | $f'(x_n)$ | $x_{n+1}$ |
|---|---|---|---|---|
| 0 | 1 | 1.7148 | -8.3254 | 1.2059 |
| 1 | 1.2059 | -0.4234 | -12.9955 | 1.1733 |
| 2 | 1.1733 | -0.0161 | -12.0119 | 1.1719 |
| 3 | 1.1719 | ±0.0006 | -11.9723 | 1.1719 |

$f(1) = 1.7148$

$$\boxed{f'(x) = -2\sin e^{\cos x} - 2x e^{x^2}}$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$= 1 - \frac{1.7148}{-8.3254}$$

$$= 1.2059$$

$f'(1) = -8.3254$

$=$

$f(1.2059) = -0.4234$

$f'(1.2059) = -12.9955$

$$x_{i+1} = 1.2059 - \frac{0.4234}{12.9993}$$

$$= 1.1733$$

~~$f(1.1712) = 9.0628 \times 10^{-3}$~~

$f(1.1733) = -0.0161$

$f'(1.1733) = -12.0119$

0.00006 88892

Read each of the questions carefully and answer showing all steps of your solution

Final Answers with no clear solution get ZERO

All the computations must be rounded to exactly four decimal places

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08

Q2- Use the **Golden Section Search** to find the **minimum** of the function (15 marks)

$f(x) = 0.2x^3 - 3\cos(x)$. Assume that $x_l = -1.5$, $x_U = 1$. Perform 4 iterations.

| $n$ | $x_l$ | $x_U$ | $d$ | $x_1$ | $x_2$ | $f(x_1)$ | $f(x_2)$ |
|---|---|---|---|---|---|---|---|
| 1 | -1.5 | 1 | 1.5451 | 0.0451 | 0.5451 | -2.9969 | -2.5328 |
| 2 | 0.5451 | 1 | 0.2811 | 0.8262 | 0.7189 | -1.9202 | -2.1833 |
| 3 | 0.5451 | 0.7189 | 0.1074 | 0.6525 | 0.6115 | -2.3281 | -2.4106 |
| 4 | 0.5451 | 0.6115 | 0.0410 | 0.5861 | 0.5705 | -2.4590 | -2.4877 |

$\phi = 0.61803$

$d = \phi\,(x_u - x_L)$

$d_1 = 1.5451$

$x_1 = x_L + d \rightarrow$

$x_2 = x_u - d$

$\boxed{x_2 = x_L}$

$x_u = x_u$

$f(x_1) > f(x_2)$

$\Rightarrow x_2 = x_u$

$\boxed{n=1}$

$d = (0.61803)(1 + 1.5)$

$= 1.5451$

$x_1 = (-1.5 + 1.5451) = 0.0451$

$x_2 = (1 - 1.5451) = 0.5451$

$\boxed{n=2}$ $d = \phi\,(\cancel{0.5451})$

$d = \phi\,(1 - 0.5451)$

$= 0.2811$

$x_1 = 0.5451 + 0.2811$

$= 0.8262$

$x_2 = 1 - 0.2811$

$= 0.7189$

$\boxed{n=3}$ $d = \phi\,(0.7189 - 0.5451)$

$= 0.1074$

$x_1 = 0.5451 + 0.1074$

$= 0.6525$

$x_2 = 0.7189 - 0.107$

$\boxed{n=4}$ $d = \phi\,(0.6113 - 0.5451) = 0.6115$

$= 0.0410$

$x_1 = 0.5861$

$x_2 = 0.5705$